

# Anomaly Detection in Social Networks: A Taxonomy of Methods, Research Challenges, and Future Directions

Fahimeh Dabaghi-Zarandi <sup>1\*</sup>, Shamim Gharaei <sup>2</sup>, Ali Zeynali <sup>3</sup>

1. Assistant Professor, Department of Computer Engineering, Vali-e-Asr University of Rafsanjan, Iran;  
f.dabaghi@vru.ac.ir
2. B.Sc. student, Department of Computer Engineering, Vali-e-Asr University of Rafsanjan, Iran;  
Shamim.gharaei2002@gmail.com
3. B.Sc. student, Department of Computer Engineering, Vali-e-Asr University of Rafsanjan, Iran;  
realalizeynali@gmail.com

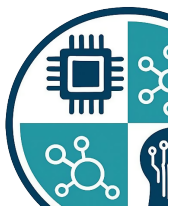
## Abstract

In today's world, social networks have become one of the most important communication platforms and are gradually turning into essential and inseparable parts of daily life. Every day, a huge amount of data is transferred through these networks; therefore, ensuring user security has become a necessity. For this reason, anomaly detection in social networks—such as identifying suspicious users and spammers—plays an important role in preventing the spread of fake information, fraud, and scams. These anomalies have different types. In this paper, we conducted a review and analysis of anomaly detection methods and proposed several categorizations for these approaches. Then, based on our proposed categorization, we evaluated and compared the methods, identified the existing challenges, and, as far as possible, suggested solutions to address them.

**Key words:** Social Network, Anomaly Detection, Anomaly Detection Classification, Similarity Measure, Evaluation Function.

## 1. Introduction

In recent decades, social networks have become one of the most important platforms for communication, information exchange, and social interaction worldwide. Platforms such as Facebook, Twitter (X), Instagram, and LinkedIn connect billions of users on a daily basis and provide the ability to share thoughts, experiences, news, and multimedia content. However, despite these opportunities, the presence of anomalous users—including spammers, fake accounts, misinformation spreaders, bots, and individuals with malicious behaviors—threatens security, trust, and the quality of user interactions. Therefore, anomaly detection in social networks has become a key and critical topic in the fields of data mining, cybersecurity, and social network analysis. An anomaly in social networks refers to users, interactions, or subgraphs whose communication, behavioral, or activity patterns deviate from the normal and expected patterns of the network. Anomaly detection is a process that identifies these deviations by analyzing network data in order to protect user security.



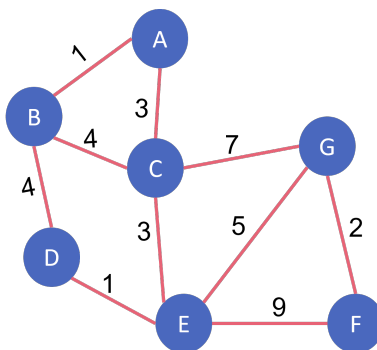
In this paper, we conducted a comprehensive review of anomaly detection methods, and various categorizations can be proposed for existing approaches in this field. Since the type of algorithm and technique used plays a more important role in the performance of anomaly detection methods, the main categorization in this study is based on the employed technique. Accordingly, we classified anomaly detection methods into four main groups: statistical methods, machine learning-based methods, graph-based methods, and hybrid methods. The details of this categorization and the corresponding methods are presented in Sections 3 and 4. In the rest of the paper, Section 2 presents the basic concepts related to anomaly detection in social networks. In Section 3, we discuss the classification of different types of anomalies and the corresponding detection methods. In Section 4 provides a review of various anomaly detection approaches. In Section 5, the methods introduced earlier are compared and evaluated, and finally, Section 6 presents a summary of the findings.

## 2. Preliminaries

In this section, we present the fundamental concepts required to understand anomaly detection in social networks. First, we provide a definition of social networks, followed by the definitions of anomalies and anomaly detection. Next, we introduce the similarity measures commonly used in anomaly detection methods. Finally, we review the evaluation metrics employed to assess and compare the performance of these methods.

### 1-2- Social Networks

Social networks are platforms that enable individuals to share their activities, interests, personal backgrounds, and real-life connections through computer-mediated communication. These networks have become popular and are now deeply integrated into everyday life [1]. From a network analysis perspective, a social network can be modeled as a graph  $G = (V, E)$ , where  $V$  represents users and  $E$  denotes social relationships or interactions between them, as shown in Figure 1. Figure 1 provides a conceptual illustration of a social network as a weighted graph, where nodes represent users and edge weights indicate the strength or frequency of interactions between them. edges may be directed or undirected and can also be weighted to represent the strength, frequency, or importance of interactions between users. For example, in an online social network, the weight of an edge might correspond to the number of messages exchanged between two users, or the duration and intensity of their interactions. Weighted edges allow for a more nuanced analysis of the network, capturing not only the existence of a relationship but also its significance. Incorporating edge weights is particularly useful in applications such as influence analysis, community detection, and anomaly detection, where the strength of connections plays a critical role in understanding network dynamics.



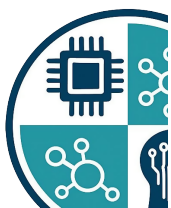


Figure 1: A conceptual illustration of a social network as a weighted graph

## 2-2- Anomaly in Social network

Anomalies are behaviors or patterns in data that do not conform to a well-defined notion of normal behavior. They are often generated by a different mechanism than the rest of the data and cannot be considered random noise.

For example, a patient's heart rate usually falls within a normal range (e.g., 60 to 100 beats per minute). If the heart rate suddenly and persistently rises to a very high value (e.g., 160) or drops to a very low value (e.g., 35), it is considered an anomaly. Such an abnormal pattern, as illustrated in Figure 2, is typically caused by a different underlying mechanism, such as cardiac arrhythmia, internal bleeding, shock, or respiratory problems.

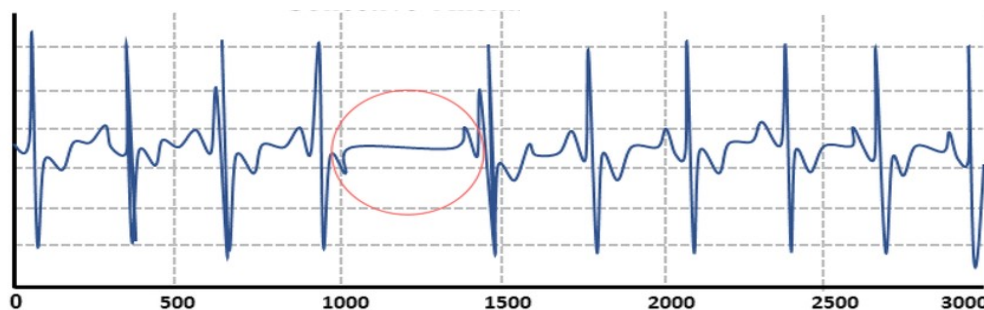
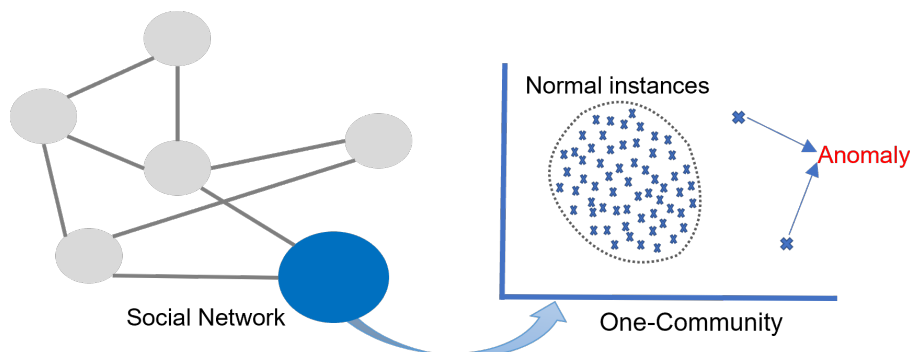
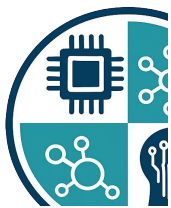


Figure 2: Anomaly in heart rate [2]

Now, regarding our field of study, anomalies in social networks refer to users, interactions, or subgraphs whose communication, behavioral, or activity patterns deviate from the normal and expected patterns of the network, as shown in Figure 3. Such deviations may include malicious activities such as spreading fake news, rumors, and misinformation, sending unsolicited messages (spam), posting advertisements containing malicious links, creating fake accounts, committing financial fraud, and other abnormal behaviors.

Online social network anomalies refer to unusual, and often illegal, user activity [1].





**Figure 3: Anomaly in Social Network**

Given the importance of anomalies in social networks, we categorized different types of anomalies in Section 3.

### ۳-2- Anomaly Detection

Anomaly detection is a process in which data, patterns, or behaviors that deviate from a defined normal behavior or pattern are identified. In social networks, anomaly detection plays a crucial role.

Anomaly detection in social networks is an important data mining task that aims to detect unexpected and suspicious users or activity [3].

These behaviors may include malicious activities such as spreading misinformation, sending unsolicited messages, creating fake accounts, financial fraud, unauthorized transactions, and other abnormal activities.

### 4-2- Similarity measures

In this section, we present the similarity measures commonly used in anomaly detection. Since anomalous instances deviate from normal patterns, they tend to have low similarity to typical behaviors. Hence, similarity computation between samples can be leveraged as a fundamental basis for anomaly scoring and detection. In this regard, similarity measures can be categorized into three types: structure-based similarity, attribute-based similarity, and structure–attribute-based similarity [4, ۵, 6].

- **Structure-based similarity**

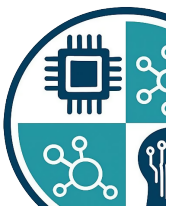
Structural-based similarity measures compute similarity between nodes by using only the topological structure of the network graph (e.g., connectivity patterns and neighborhood relations), while ignoring node attributes. These measures are widely used in community detection and graph-based anomaly detection, since anomalous nodes often exhibit unusual structural connectivity compared to normal nodes.

- **Attribute-based similarity**

Attribute-based similarity measures quantify the similarity between nodes, entities, or samples solely based on their intrinsic attributes or feature values, without considering the network topology or structural connections. By evaluating similarity based on attributes, these methods can detect anomalies that deviate from normal feature patterns, even if their structural positions or connections appear typical.

- **Structure–Attribute-based similarity**

Unlike structure-based measures, which rely solely on connectivity patterns, and attribute-based measures, which consider only node features, structure–attribute-based similarity measures combine both sources of information to provide a more accurate



similarity estimation and enhance the detection of anomalies that deviate structurally, semantically, or in both aspects.

## 5-2- Evaluation Metrics

For evaluating the performance of anomaly detection methods in social networks, standard machine learning and data mining metrics are commonly used. These metrics are generally divided into two main categories: accuracy and detection quality metrics and efficiency and operational metrics [6].

**Detection Quality Metrics:** These metrics are calculated by comparing the method's output with real or labeled data. For anomaly detection, various evaluation metrics are commonly used, including Precision, Recall, Jaccard Similarity, F1-score, AUC-ROC, and Accuracy.

- **Precision:** Precision is a measure that represents the ratio of correctly detected anomalies to the total number of detected anomalies. The formula in Equation 1.

$$Precision = \frac{True\ positives}{True\ positives + False\ positive} \quad (1)$$

If there are no positive predictions, precision is defined as 0 to avoid division by zero. This metric indicates what percentage of the cases flagged as anomalies by the model are actually true anomalies.

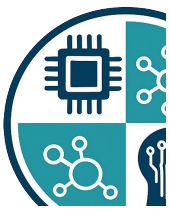
- **Recall:** The ratio of correctly detected anomalies to the total number of actual anomalies. The formula in Equation 2.

$$Recall = \frac{True\ positives}{True\ positives + False\ negatives} \quad (2)$$

If there are no actual positive cases, recall is defined as 0 to avoid division by zero. This metric shows how many percent of the true anomalies have been detected by the model.

- **Jaccard similarity:** Comparison of the set of detected anomalies with the ground-truth set. It is defined as the size of the intersection of the sets divided by the size of their union. The formula in Equation 3.

$$Jaccard\ similarity\ (A,B) = \frac{|A \cap B|}{|A \cup B|}$$



(3)

The value of Jaccard Similarity ranges from 0 to 1, where 1 indicates that the two sets are identical, and 0 indicates that they have no elements in common.

- **F1- Score:** The harmonic mean of Precision and Recall. It provides a balanced metric that combines both measures into a single one. The formula in Equation ۴.

$$F1 - Score = 2 \times \frac{Precision \times Recall}{Precision + Recall} \quad (4)$$

- **AUC–ROC:** The primary metric for evaluating a method's ability to distinguish between anomalies and normal instances. The Receiver Operating Characteristic (ROC) curve is obtained by plotting the True Positive Rate (TPR) against the False Positive Rate (FPR) under different decision thresholds. Are defined by the formulas in Equation 5 and 6.

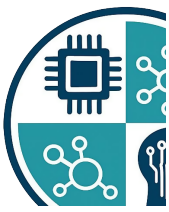
$$True\ Positive\ Rate = \frac{True\ positives}{True\ positives + False\ negatives} \quad (5)$$

$$False\ Positive\ Rate = \frac{False\ positives}{False\ positives + True\ negatives} \quad (6)$$

TP is the number of anomalies correctly detected, FN is the number of anomalies that the model failed to detect, FP is the number of normal instances incorrectly classified as anomalies, and TN is the number of normal instances correctly classified as normal. The Area Under the Curve (AUC) is defined as the area under the ROC curve and defined by the formulas in Equation 7.

$$AUC = \int_0^1 TPR(FPR) d(FPR) \quad (7)$$

The AUC value ranges from 0 to 1. A higher AUC indicates that the model has a better ability to distinguish anomalous instances from normal ones. An AUC of 1 represents perfect discrimination, while an AUC of 0.5 indicates performance equivalent to random.



- **Accuracy:** It is one of the basic and simple metrics for evaluating model performance. This metric indicates the overall percentage of instances that the model has correctly detected. The formula in Equation 8.

$$Accuracy = \frac{True\ positives + True\ negatives}{True\ positives + True\ negatives + False\ positives + False\ negatives} \quad (8)$$

**Efficiency & Operational Metrics:** These metrics are for evaluating the practical performance and applicability. These metrics include, Runtime/ Computational Time, Scalability, and Modularity.

- **Runtime/ Computational Time:** This metric refers to the amount of time required for a method to complete its execution on a given dataset. It reflects the computational efficiency of the approach and is especially important when dealing with large-scale social networks.
- **Scalability:** This metric refers to the ability of a method to efficiently handle large-scale graphs with a high number of nodes and edges. A scalable anomaly detection approach should maintain acceptable computational cost and memory usage as the size of the social network grows. Scalability is particularly important in real-world social networks, where graphs can be massive and continuously evolving.
- **Modularity:** A metric used to evaluate the quality of detected communities in community-based methods.

### 3. Categorization of Anomalies and Detection Approaches

In this section, we aim to categorize different types of anomalies. We also provide a categorization of the existing methods for anomaly detection.

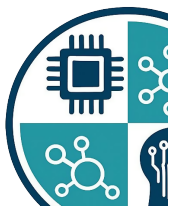
#### 1-3- Categorization of Anomalies in Social Networks

Based on the studies we conducted, anomalies in social networks are categorized into four distinct types: Point, Contextual, Collective, and Structural anomalies, which are discussed below.

- **Point Anomalies**

A Point anomaly refers to an individual user, post, or node that significantly deviates from the normal pattern of other users or data, while the rest of the data behaves normally.

An example is an Instagram account that posts thousands of identical comments (e.g., advertisements) under different posts in a single day, while normal users typically leave only 20–30 comments per day.



- **Contextual Anomalies**

A contextual anomaly refers to behavior that appears normal across the network as a whole but is unusual within a specific context, such as the community or group to which the user belongs.

An example in a Telegram group for "university students," a person who is 60 years old posts extreme political content. This behavior may be normal on Telegram overall, but it is completely unusual within the 18–25-year-old student community.

- **Collective Anomalies**

A collective anomaly refers to a group of users, posts, or subgraphs that may appear normal individually, but together exhibit an unusual pattern.

An example on Twitter (X), a group of 50 accounts simultaneously and coordinately repost a fake news item (e.g., "So-and-so politician resigned") and trend the same hashtag. Each account alone may seem normal, but this collective coordination is unusual.

- **Structural Anomalies**

A structural anomaly refers to deviations in the network's structure its relationships and connections regardless of individual attributes. Examples include bridge nodes, pendant nodes, or unusual community formations.

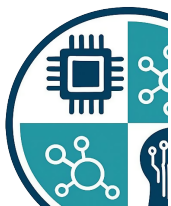
An example a LinkedIn account, which is an ordinary employee, connects to 800 senior managers across various industries (banking, oil, technology, healthcare) without any logical reason. This account acts as a bridge between different communities and may be attempting to gather information or engage in fraud.

## 2-3- Algorithm-Based Classification of Anomaly Detection Methods

So far, numerous methods have been proposed for anomaly detection. These methods can be categorized based on the type of similarity measure used by the algorithms (as discussed in Section 2-4) or based on the type of evaluation performed (as described in Section 2-5). However, given the importance of the techniques employed by the algorithms for application domains and future research, our focus is on the type of algorithm used for anomaly detection. Accordingly, we classify the methods into four main categories: statistical-based methods, machine learning-based methods, graph-based methods, and hybrid methods, which are discussed below.

### 1-2-3- Statistical Based Methods

Statistical methods are common approaches for anomaly detection in social networks. They assume that normal data follow a certain statistical pattern, and anomalies are data points that deviate clearly from this pattern. These methods use basic statistical measures such as mean, variance, local density, and probability distributions. Although they are simple and easy to interpret, statistical methods have limitations when dealing with high-dimensional, non-stationary, and complex network data, and are mainly suitable for detecting large-scale or global anomalies.



In these methods, for example, the density or distance of each point (user) is compared with that of its neighboring points. If the density of a point is significantly lower than the average density of its neighbors, or if its distance from the neighbors is unusually large, it is detected as an anomaly. These methods usually operate in an unsupervised manner and rely only on local features.

### **2-2-3- Machine Learning Based Methods**

Machine learning-based methods for anomaly detection first learn normal behavior patterns from the data and identify instances that significantly deviate from these patterns as anomalies. The model's output is usually a score or label indicating how far each instance deviates from normal behavior. These methods can be applied in supervised, semi-supervised, or unsupervised settings. In social networks, they are used to analyze user features, relationships, and behavioral patterns. Machine learning models are capable of detecting complex and non-linear patterns and identifying unusual instances, which has led to their widespread use in anomaly detection studies.

Supervised methods require labeled data (normal and anomalous). In contrast, unsupervised methods learn the normal patterns from data and detect anything that deviates from the learned pattern as an anomaly. However, these methods may be sensitive to noise.

### **3-2-3- Graph Based Methods**

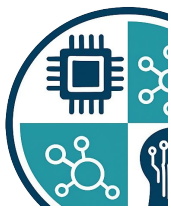
Graph-based methods detect anomalies by modeling social networks as graphs, where nodes represent users and edges represent relationships or interactions. These methods analyze the network's structure and connectivity patterns to identify unusual nodes or subgraphs. Anomalies may include nodes with abnormal behavior, incoherent communities, or sudden structural changes. The main advantage of these methods is their ability to capture complex relationships and detect unusual structural and behavioral patterns. These methods are particularly well-suited for relational data, such as social networks.

### **4-2-3- Hybrid Methods**

Hybrid methods for anomaly detection in social networks are designed to simultaneously leverage the strengths of multiple approaches. These methods typically integrate two or more categories of techniques, including statistical methods, machine learning, graph-based approaches, and deep learning, to mitigate the limitations of individual methods. Given that social network data are inherently multi-dimensional—encompassing structural, behavioral, and temporal characteristics—single-method approaches are often limited in their ability to model all these aspects simultaneously.

## **4. Anomaly Detection Methods**

In this section, we present several anomaly detection methods in social networks, based on our identified categories in Section 3.



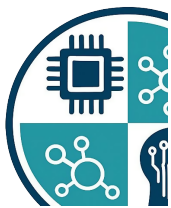
#### 1-4- Statistical Based Methods

The Relative Density-based Outlier Factor (RDOF) is a density-based method designed to detect local outliers [7]. For each data point, the  $k$  nearest neighbors are first identified, and the  $k$ -th neighbor is selected as a reference point. The distance between the target point and the reference point is then computed, and a local neighborhood is formed by combining the nearest neighbors of the target point with the mutual neighbors of the reference point. Next, the local density of the target point is calculated based on the number of neighbors and their distances, and the density of the reference point is computed in a similar way. The RDOF value is defined as the ratio of the reference point density to the density of the target point. Larger values indicate a higher likelihood of being an outlier, while values close to one usually correspond to normal points. This procedure is repeated for all data points, and those with the highest RDOF values are reported as outliers.

Local density feedback outlier factor (LDF) is a density-based outlier detection method that aims to address the limitations of earlier approaches such as LOF and RDOF by introducing a density feedback mechanism [8]. First, the data are reduced using Principal Components Analysis (PCA) [9] to mitigate correlations and reduce noise. Then, Natural Neighbor Search with a Ball-Tree [10] structure is employed to automatically determine the number of neighbors ( $k$ ) based on the intrinsic data distribution. After computing the initial local density of each point, the density feedback mechanism is applied, where density differences within the neighborhood are treated as feedback signals. Based on the neighborhood similarity principle, which assumes that neighboring points should exhibit similar density values, the local densities are iteratively updated over multiple rounds. In each iteration, the density feedback value of each point is refined according to its neighbors. Through repeated updates, more global information is gradually propagated through chains of neighborhoods and injected into the local density estimation. As a result, low-density normal points tend to be pulled toward the density level of surrounding normal points, while outliers typically undergo smaller changes. After convergence, the Feedback Outlier Factor (FOF) is defined, usually as the inverse of the feedback-refined density (or similarly to LOF but computed using the adjusted density). Finally, the points with the highest FOF values are identified as outliers.

#### 2-4- Machine Learning Based Methods

The Density Peak-guided K-means (DPK) method operates in two stages [11]. In the first stage, the data are divided into natural groups (clusters). Instead of randomly selecting the initial centroids in K-means, an improved version of the Density Peaks Clustering (DPC) method is used. In this method, the density of points is computed using K-Nearest Neighbors ( $k$ -NN) and kernel density, rather than a fixed value. As a result, suitable central points are selected as the initial centroids, and then the K-means algorithm is applied to perform clustering. In the second stage, anomalies are detected within each cluster. For each point, the distance from the cluster center is calculated, and the points are sorted based on this value. Then, a line or curve is fitted using the least squares method, and the slope of this line/curve is examined for each point. Points that, in addition to being far from the center, cause a sudden increase in the slope receive a higher anomaly score and are identified as anomalies.



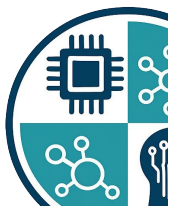
The Multimodal Granular Distance-based Outlier Detection (MGDOD) method is an unsupervised anomaly detection algorithm specifically designed for highly heterogeneous multimodal data [12]. The main idea is that, instead of directly comparing raw features (which may differ in scale, distribution, and type), the data are transformed into a shared granular space. To achieve this, a small reference set of samples is first selected. For each sample and each modality, its similarity to the reference samples is calculated, producing a short granular vector. Two types of granular distance are then defined: one based on the intersection-over-union ratio, and the other based on the mean symmetric differences (XOR). Finally, for each sample, the average granular distance to all other samples is computed, called the Expected Outlier Factor (EOF). Samples with high EOF values are identified as anomalies.

MGDOD method, by reducing the effect of modality heterogeneity, providing good interpretability, and showing relatively high robustness against noise and dispersion, has demonstrated competitive or superior performance compared to classical distance-based and density-based methods.

### 3-4- Graph Based Methods

The Graph Attention Network (GAT) + Graph Autoencoder (GAE) method operates as follow [13]. In this approach, the social network is modeled as an undirected graph, where nodes represent users and edges denote relationships or interactions between them. In the first stage, a Graph Attention Network (GAT) is employed to aggregate each user's features with information from its neighbors. Unlike simpler methods that treat all neighbors equally, GAT learns to assign higher weights to more relevant neighbors, enabling the construction of more informative and accurate node representations. The output of this stage is a compact latent representation (embedding) for each node, which preserves key information about both node attributes and relational structure. These embeddings are then fed into the decoder of a Graph Autoencoder (GAE). The decoder aims to reconstruct the original graph structure by predicting which node pairs should be connected, typically using the inner product of node embeddings. The model is trained in an unsupervised manner by minimizing the reconstruction error, defined as the difference between the original graph and the reconstructed one. Nodes with high reconstruction errors are identified as anomalous, as their connectivity patterns and behaviors deviate from the dominant network structure. By combining GAT and GAE, the method effectively captures important local relationships while also modeling the global structure of the network. Experimental results on Facebook data demonstrate superior performance in terms of AUC and F1 compared to baseline models, highlighting its suitability for financial applications such as fraud detection.

The Spectral Filter (SpecF) method operates as follows [14]. First, it detects the network communities or receives them in advance. Then, it expands the adjacency matrix to incorporate the community structure; in this way, nodes within the same community (even if not directly connected) gain a weak connection and receive higher weights compared to nodes outside the community. Using this new matrix, the graph Laplacian is constructed. Next, node features (signals) are transformed to the frequency domain using the graph Fourier transform. The cutoff frequency of the low-pass filter is set approximately to the k-th smallest eigenvalue



( $k$  = number of communities). The low-pass filter is applied to attenuate sharp and sudden changes (high frequencies). The filtered signal is then transformed back to the graph domain using the inverse transform. The residual (difference) between the original signal and the filtered signal is computed. For each individual community, a local threshold is set as the mean residual plus two standard deviations. Nodes whose residuals exceed their community threshold are considered anomalies.

Since this method does not have a specific name in the original study, for convenience, we refer to it as Community Boundary and Outlier Detection (CBOD) in the remainder of this paper. The CBOD method presents a fast, community-structure-based approach for anomaly detection in social networks [15]. This method, instead of examining all nodes, focuses only on pendant nodes (degree = 1) and community boundary nodes (CBNs). First, the social network is modeled as an undirected graph  $G(V, E)$ . Then, using community detection algorithms such as Louvain or Girvan–Newman (GN), disjoint communities are identified.

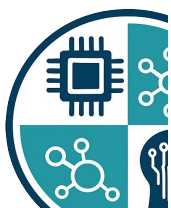
Only two groups of nodes are considered for anomaly detection: pendant nodes (degree = 1), because they have very weak connections to their own community, and community boundary nodes (CBNs), which are nodes that have at least one edge to another community (inter-community edge). Fully internal nodes within communities (with high degree and no external edges) are not processed at all, which significantly reduces computational complexity.

Then, for each boundary node, the anomaly score is calculated using the permanence metric. This metric quantitatively expresses the node's belonging to its main community. Nodes whose permanence score is less than or equal to a threshold are detected as anomalies. Pendant nodes are considered anomalous directly without calculating the score.

#### 4-4- Hybrid Methods

The Community Anomaly detection (CANom) method, computes a numerical anomaly score for each node instead of performing binary classification, ranking them accordingly [3]. Assuming the locality of anomalies, the CANom method operates in two stages. First, the network is divided into disjoint communities using the Louvain algorithm. Then, the anomaly score of each node is calculated based on the combination of two components: a structural component and an attribute-based component. The structural component measures the importance and connectivity of a node within its community, such that weak connections or links to low-importance nodes lead to an increased anomaly score. In the attribute-based component, which is the main novelty of the method, the most important attributes are first selected through a ranking method, and then the deviation of each node from the average of its community's attributes is computed. Finally, these two components are combined using adjustable weights, and a final anomaly score is obtained for ranking nodes.

The Decision Tree (C5.0), Support Vector Machine (SVM) and Naïve Bayesian classifier (NBC) (DT-SVMNB) method is a hybrid approach whose primary objective is to identify anomalous, depressed, and particularly suicide-prone users in social networks, with a specific focus on Facebook [1]. The method employs three machine learning algorithms in a cascading manner. Two main categories of features are extracted: profile-based features and content-



based features. In the first stage, the entire social network dataset is fed into the C5.0 decision tree algorithm, which classifies users into normal and anomalous groups. In the second stage, only the users identified as anomalous in the previous step are passed to a Support Vector Machine (SVM) with a Gaussian kernel, where they are further categorized into happy and disappointed/depressed users. The SVM is selected at this stage due to its high accuracy in binary classification tasks. In the third stage, only the disappointed/depressed users are processed by a Naïve Bayes Classifier (NBC). Using a suicidal word dictionary and the computation of conditional probabilities, this classifier distinguishes users with suicidal ideation from non-suicidal users. Naïve Bayes is considered suitable for this final stage due to its computational simplicity and robust performance on smaller datasets.

The Temporal Knowledge Graph + Graph Neural Network (TKG-GNN) method is designed based on a Temporal Knowledge Graph of User Behavior and graph representation optimization [16]. This method operates in two main stages. First, users' behavioral logs are transformed into a temporal knowledge graph by extracting entities and relations (using NER and regular-expression rules) and performing entity alignment based on cosine similarity. Then, by leveraging a dual graph embedding technique (network topology + temporal information) and a local-global hybrid attention mechanism (Hybrid Attention), the graph is completed and missing relations are predicted (ATKGC model).

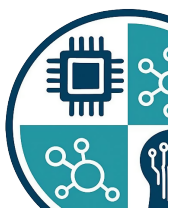
In the second stage, temporal subgraphs are encoded into vector representations using two Graph Convolutional Networks (GCNs): one applied to the original graph to capture users' semantic features, and the other applied to the line graph to capture interaction features. These representations are combined with an LSTM to model temporal dynamics, and finally an anomaly score is computed using an Encoder-Decoder-Encoder architecture. If the anomaly score of a subgraph exceeds a predefined threshold, it is identified as anomalous behavior.

## 5. Comparison of Proposed Methods

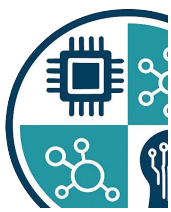
In this section, we compare all the methods introduced in Section 4. To facilitate better understanding, we also provide a table. Table (1) summarizes the comparison of the proposed methods.

**Table 1: Comparison of anomaly detection methods**

| Titel & year   | Method summary                                                                                                                                                                                   | Anomaly Type          | Methodology | Similarity Measure | Evaluation Function           |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|-------------|--------------------|-------------------------------|
| RDOF [7], 2022 | Aims to accurately detect local outliers in datasets with varying neighborhood densities by proposing Relative Density-based Outlier Factor (RDOF) using mutual proximity and influential space. | Point, and contextual | Statistical | Structure-based    | Recall, Precision, Rank Power |



|                     |                                                                                                                                                                                                                             |                                   |                  |                           |                                        |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|------------------|---------------------------|----------------------------------------|
| LDF [8],<br>2024    | Aims to improve accuracy in density-based outlier detection by incorporating neighborhood similarity and global information via iterative density refinement with feedback mechanism.                                       | Point                             | Statistical      | Structure-based           | Precision, AUC                         |
| DPK [11],<br>2025   | Aims to improve robustness in detecting global/local outliers and sparse clusters by integrating optimized Density Peaks Clustering (DPC) with K-means objective function and derivative-based anomaly scoring.             | Point, collective, and contextual | Machine Learning | Structure-based           | AUC, Precision, F1-Score, Recall       |
| MGDOD [12], 2025    | Aims to handle complex, sparse, nonlinear, and multimodal data for robust outlier detection by introducing Multimodal Granular Distance-based Outlier Detection (MGDOD) with granular computing and multimodal granulation. | Point, and contextual             | Machine Learning | Structure-Attribute-based | AUC<br>F1-Score<br>Precision<br>Recall |
| GAT+GAE [13], 2025  | Aims to enhance abnormal user detection and in social networks by combining Graph Autoencoders (GAE) and Graph Attention Networks (GAT) for dynamic neighbor aggregation and reconstruction error-based anomaly scoring.    | Structural, and contextual        | Graph Based      | Structure-based           | AUC<br>F1-Score<br>Precision<br>Recall |
| SpecF [14],<br>2022 | Aims to detect contextual anomalies in attributed networks with community structure by proposing spectral graph filtering (SpecF) that incorporates community information into the Laplacian matrix and cutoff frequency.   | Structural, and contextual        | Graph Based      | Structure-based           | Accuracy, AUC                          |
| CBOD [15], 2023     | Aims to identify outliers from a global perspective in social networks by focusing on pendant nodes and community boundary nodes (CBNs) using permanence metric, reducing search space for efficiency.                      | Structural, and collective        | Graph Based      | Structure-based           | F1-Score, Jaccard similarity           |



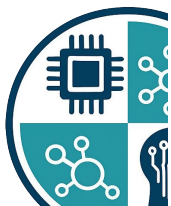
|                           |                                                                                                                                                                            |                                                 |        |                                   |                                                |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|--------|-----------------------------------|------------------------------------------------|
| CAnom<br>[3], 2022        | Aims to detect and rank anomalous users in OSN communities by measuring structural and attribute deviations from normal behavior using a community-based approach (CAnom). | Structural,<br>point, and<br>contextual         | Hybrid | Structure–<br>Attribute-<br>based | Accuracy,<br>AUC-ROC                           |
| DT-<br>SVMNB<br>[1], 2021 | Aims to classify normal/abnormal users in social networks with high accuracy by cascading Decision Tree (C5.0), SVM, and Naïve Bayes on profile/content features.          | point, and<br>contextual                        | Hybrid | Attribute-<br>based               | Accuracy,<br>Precision,<br>Recall,<br>F1-Score |
| TKG-GNN<br>[16], 2024     | Aims to detect anomalous behaviors by constructing temporal knowledge graph from user logs and using optimized graph embedding with hybrid attention and GNN.              | Structural,<br>Contextual,<br>and<br>Collective | Hybrid | Structure-<br>based               | F1-Score                                       |

As shown in Table 1, each row corresponds to a single method. The first column lists the method name (as presented in *Section 3*) along with its year of publication. The second column provides a brief summary of the method. The third column indicates the type of anomaly detected or addressed by the method (as described in *Section 3-1*). The fourth column specifies the method category based on the classification proposed in *Section 3-2*. The fifth column lists the similarity measures used by the method (as discussed in *Section 1-4*). Finally, the last column presents the evaluation metrics employed for each method, based on *Section 1-5*.

Overall, recent methods have demonstrated significant improvements in detection accuracy. Simpler approaches, such as RDOF, are fast, straightforward, and achieve high accuracy on small datasets; however, they face limitations when applied to large or heterogeneous data. DPK methods are robust against scattered clusters and provide precise scoring, but they are computationally more intensive compared to simpler approaches. GAT+GAE achieves the best performance on large, real-world datasets, yet it is computationally demanding and requires the use of a graphics processing unit (GPU). CAnom shows resilience to an increasing number of features, but it requires careful tuning of the parameter  $\alpha$  and appropriate feature selection.

In summary, while each method offers distinct advantages, none simultaneously optimizes accuracy, scalability, and computational efficiency. Therefore, selecting an appropriate approach depends on the characteristics of the dataset and the available computational resources.

To facilitate the work of readers who intend to conduct research in this area, we list some of the main challenges in the following.



### 1-5- Challenges in Anomaly detection

Despite the progress and research in anomaly detection in social networks, this field still faces many theoretical and practical challenges. Some of these challenges include Scalability issues, Data Heterogeneity, Contextuality, and Label Scarcity, which are discussed below.

- **Scalability:**

Scalability means the ability of a method to maintain computational performance (time and memory) with increasing data size. Since social networks generate billions of interactions daily and their data are continuously changing, anomaly detection methods must be able to process very large networks in real-time or near real-time.

To address this challenge, techniques such as reducing the search space and focusing on key nodes, processing subgraphs or graph sampling, and employing distributed and parallel methods can be applied.

- **Data Heterogeneity:**

Data come in different types (numerical, textual, visual, and structured or semi-structured). Social networks generate these data in various structures; therefore, anomaly detection methods must be able to process and analyze such diverse data simultaneously.

Several approaches have been proposed to address this issue. Multi-modal models enable simultaneous processing of different data types, while feature fusion techniques combine multiple features prior to analysis. Knowledge graphs preserve relationships among heterogeneous data and model both context and structure. Moreover, attributed or heterogeneous graph embedding algorithms transform nodes and their features into numerical vectors, facilitating better anomaly detection. Semi-supervised and unsupervised methods are also effective when labeled data is scarce, and preprocessing and standardization help reduce noise and harmonize feature scales.

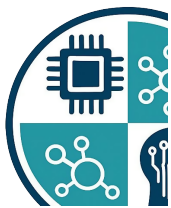
- **Contextuality:**

A behavior may appear normal at the overall network level but be anomalous within a specific local community or subnetwork. This dependence on local context and structure makes anomaly detection highly challenging. Moreover, many anomalies manifest in a stealthy and sophisticated manner, such as advanced bots, making them difficult to detect.

Therefore, community-based methods that combine network structure and node attributes can better model local patterns and improve the detection accuracy of such anomalies.

- **Label Scarcity:**

Labeled data scarcity is one of the major challenges in anomaly detection. In social networks, the number of labeled anomalous samples is very limited, and manual labeling is costly and time-consuming. This limitation makes the use of supervised



learning methods difficult and often requires unsupervised or semi-supervised algorithms.

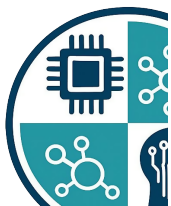
## ۵. Conclusion

As discussed, this paper addressed the problem of anomaly detection in social networks. First, the fundamental concepts were introduced, including the definition of social networks, anomalies, and anomaly detection. Then, a categorization of anomaly types was presented, where anomalies were divided into four groups: point, collective, contextual, and structural anomalies. In addition, anomaly detection methods were categorized into four classes: statistical approaches, machine learning methods, graph-based methods, and hybrid approaches. Next, based on the proposed categorizations, the related methods were reviewed and compared. As shown in Table 1, this comparison led to the identification of several key challenges, including scalability, data heterogeneity, contextual dependency, and the lack of labeled data. These challenges were discussed in Section 2-5, and, as far as possible, potential solutions were proposed to address them.

This study, by providing a classification and analytical framework, can be used as a fundamental reference for the selection and development of novel anomaly detection methods in social networks.

## References

- [1] Rahman, Md Shafiur, Sajal Halder, Md Ashraf Uddin, and Uzzal Kumar Acharjee. "An efficient hybrid system for anomaly detection in social networks." *Cybersecurity* 4, no. 1 (2021): 10
- [2] Al-Amri, Redhwan, Raja Kumar Murugesan, Mustafa Man, Alaa Fareed Abdulateef, Mohammed A. Al-Sharafi, and Ammar Ahmed Alkahtani. "A review of machine learning and deep learning techniques for anomaly detection in IoT data." *Applied Sciences* 11, no. 12 (2021): 5320.
- [3] Zardi, Hedia, Hanen Karamti, Walid Karamti, and Norah Saleh Alghamdi. "Detecting anomalies in network communities based on structural and attribute deviation." *Applied Sciences* 12, no. 22 (2022): 11791.
- [4] Dabaghi-Zarandi, Fahimeh, and Parsa KamaliPour. "Community detection in complex network based on an improved random algorithm using local and global network information." *Journal of Network and Computer Applications* 206 (2022): 103492.
- [5] Dabaghi-Zarandi, Fahimeh, Mohammad Mehdi Afkhami, and Mohammad Hossein Ashoori. "Community detection method based on random walk and multi objective evolutionary algorithm in complex networks." *Journal of Network and Computer Applications* 234 (2025): 104070.
- [6] Dabaghi-Zarandi, Fahimeh, Pouria HosseiniDokht, Asieh Ghanbarpour, and Zohreh Davarzani. "Community detection in complex networks using vertex cover and an adaptive community transformer engine." *Chaos, Solitons & Fractals* 204 (2026): 117714.
- [7] Wahid, Abdul, and Annavarapu Chandra Sekhara Rao. "RDOF: An outlier detection algorithm based on relative density." *Expert Systems* 39, no. 2 (2022): e12859.
- [8] Zhang, Zhongping, Yuehan Hou, Yin Jia, and Ruibo Zhang. "An Outlier Detection Algorithm Based on Local Density Feedback Outlier Factor." (2024).



- [9] Souza, Tancredo. "Principal Component Analysis (PCA)." In *Advanced Statistical Analysis for Soil Scientists*, pp. 43-56. Cham: Springer Nature Switzerland, 2025.
- [10] Zhang, Lindan, Guangjie Wang, Li Peng, Wenfu Peng, and Ji Zhang. "Applying pareto frontier theory and ball tree algorithms to optimize growth boundaries for sustainable mountain cities." *Journal of Urban Management* 14, no. 2 (2025): 468-484.
- [11] Xia, Hao, Yu Zhou, Jiguang Li, Lei Bai, Jichun Li, and Fengyu Zhou. "Outlier detection via optimized density peaks clustering and K-means-derived objective function." *Chaos, Solitons & Fractals* 199 (2025): 116791.
- [12] Huang, Tiancai, Shiwang Zhang, Hao Luo, Jinsong Lyu, Ying Zhou, and Yumin Chen. "An Outlier Detection Algorithm Based on Multimodal Granular Distances." *Mathematics* 13, no. 17 (2025): 2812.
- [13] Zhang, Yiwei. "Social network user profiling for anomaly detection based on graph neural networks." In *2025 5th International Conference on Artificial Intelligence and Industrial Technology Applications (AIITA)*, pp. 1197-1201. IEEE, 2025.
- [14] Francisquini, Rodrigo, Ana Carolina Lorena, and Mariá CV Nascimento. "Community-based anomaly detection using spectral graph filtering." *Applied Soft Computing* 118 (2022): 108489.
- [15] Dey, Arnab, B. Rushi Kumar, Bishakha Das, and Arnab Kumar Ghoshal. "Outlier detection in social networks leveraging community structure." *Information Sciences* 634 (2023): 578-586.
- [16] Xing, Ling, Shiyu Li, Qi Zhang, Honghai Wu, Huahong Ma, and Xiaohui Zhang. "Anomalous behavior detection based on optimized graph embedding representation in social networks." *Journal of King Saud University-Computer and Information Sciences* 36, no. 7 (2024): 102158.